

Arunakiri Venkatachalam

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PROFESSIONAL SUMMARY

Data Scientist with 8+ years of experience building end-to-end AI and data engineering solutions across the automotive and pharmaceutical industries. Specialises in scalable ETL pipelines, LLM-powered enterprise tools, and cloud-native applications. Proven track record of delivering quantified impact: 30% reduction in third-party software licensing costs, ~60% reduction in document-search time, and ~80% OCR accuracy on unstructured regulatory data.

TECHNICAL SKILLS

Languages & Frameworks: Python, Django, Shell Scripting, REST APIs, SQLAlchemy, SQL

AI / Machine Learning: Machine Learning, Deep Learning, NLP, LLM Agents, RAG Pipelines, OCR, LangChain, Vector Databases

Data Engineering: ETL Pipeline Design, PostgreSQL, Apache Iceberg, AWS S3, High-Frequency Data Processing

DevOps & Cloud: Docker, Kubernetes, CI/CD, Git / GitHub, Keycloak, OAuth 2.0, On-Premises Secure Environments

Analytics & Visualisation: Power BI, Grafana, Databricks, Plotly Dash, Streamlit, KPI Dashboard Design

Integrations: MCP (Model Context Protocol), Conversational Analytics, Retrieval-Augmented Generation

PROFESSIONAL EXPERIENCE

Data Scientist · Volvo Group

October 2023 – Present

- Designed and maintained scalable data solutions on a secure on-premises Data Science Lab supporting 200+ concurrent users across engineering and analytics teams.
- Built high-frequency ETL pipelines and interactive Plotly Dash web applications for hydrogen engine test rigs and fleet data streams, reducing pipeline latency by over 40%.
- Developed a production-grade LLM agent system using RAG pipelines and LangChain, enabling natural-language querying of proprietary engineering documents and reducing document-search time by approximately 60%.
- Integrated MCP (Model Context Protocol) connectors to allow the agent to query live PostgreSQL databases and render charts directly within the chat interface.
- Built real-time monitoring dashboards in Grafana and Databricks, delivering operational visibility across multiple hydrogen engine test rigs for 3 engineering teams.
- Delivered a full-stack access-management portal (Django + PostgreSQL) that automated Data Science Lab user provisioning and resource allocation, eliminating manual ticketing.
- Led the Volvo Data Science Conference 2026, coordinating cross-functional sessions and knowledge-sharing tracks across 10+ teams site-wide.

Lead Analyst · HCL Technologies

October 2021 – October 2023

- Delivered end-to-end data engineering solutions as a consultant embedded in Volvo's internal teams, spanning multiple business units and analytical programmes.
- Drove the MATLAB-to-Python migration initiative, cutting third-party software licensing costs by 30% while improving solution reproducibility and long-term maintainability.
- Re-engineered three core analytical solutions — tire-wear modelling, geo-location tracking, and battery state-of-charge estimation — as robust, production-ready Python implementations.
- Designed reusable Python libraries and data-processing modules that standardised workflows and accelerated delivery velocity across 5+ parallel projects.
- Conducted structured Python training sessions for 20+ engineers and analysts, driving measurable adoption of programming-based analytics across Volvo teams.

Senior Systems Engineer · Infosys

April 2018 – October 2021

- Built OCR-based data extraction pipelines for Novartis (pharmaceutical domain), achieving ~80% extraction accuracy on complex multi-format regulatory documents.
- Automated structured data extraction from unstructured PDFs and scanned images, reducing manual data-entry effort by over 70% for compliance-critical workflows.
- Designed a PostgreSQL data storage architecture to centralise extracted product information, enabling efficient querying and downstream reporting for QA and business teams.
- Developed Power BI dashboards that provided business users with self-service access to product information and quality metrics, replacing a manual reporting cycle.
- Collaborated with QA teams to define validation rules and ensure regulatory compliance in all data outputs, supporting audit readiness across multiple product lines.

EDUCATION

M.Tech in Data Science & Engineering · BITS Pilani, Hyderabad | March 2020 – August 2022

Coursework: Machine Learning, Deep Learning, Big Data Analytics, Statistical Modelling, NLP

B.E. in Electronics & Communications · CEG, Anna University, Chennai | August 2012 – November 2016

Foundation in signal processing, embedded systems, and communication networks